

SUPERPOSITION OF MARKOV RENEWAL PROCESSES

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SUMMARY

Some properties and applications of the resulting process when N independent Markov Renewal Processes, in equilibrium, are superimposed, are considered in this paper. The interval between two successive visits to a particular State, the asymptotic variance (as the time $t \rightarrow \infty$) of the number of visits to a State, in the superposed process and the limiting distribution of the number of visits to a particular State when N also tends to infinity are considered. Applications to a human reproduction model and predator-prey model are considered.

1. Introduction

Consider an ordinary renewal process, where the cumulative distribution function (c.d.f.) of the interval between any two successive renewals is $G(x)$. If such a process is observed after a long time has elapsed, it is called an equilibrium renewal process (see Cox (1962)). When several such equilibrium processes are superimposed, Cox and Smith (1954) have given a number of interesting results for the superposed process. It is the purpose of this paper to extend these results to Markov Renewal Processes (M.R.P.).

In order to do this, it is first necessary to review Cox's results. We shall denote by $g(s)$ the Laplace-Stieltjes Transform (L.-S.T.) of $G(x)$, i.e.,

$$(1.1) \quad g(s) = \int_0^{\infty} e^{-sx} dG(x).$$

In general, we shall use the corresponding lower case letter for denoting the L.-S.T. of a function. Let $\mu^* = \mu^*_1, \mu^*_2, \mu^*_3$, be the first three moments (about zero) of the distribution $G(x)$ and let C be the coefficient of variation. Then Cox and Smith (1954) have shown that, when N independent, identically distributed equilibrium renewal processes are superimposed, the probability density function (p.d.f.) of the interval between suc-

cessive renewals of the superposed process is

$$(1.2) \quad -\frac{d}{dy} \{ (1 - G(y)) \left[\int_y^\infty \frac{1 - G(x)}{\mu^*} dx \right] \}^{N-1}$$

They also show that the variance of the number of renewals in an interval $(0, t)$ for the superposed process, $V_p(t)$, is asymptotically (for large t)

$$C^2 t \lambda$$

where $\lambda^{-1} = \mu^*/N$ (the mean interval between events in the pooled output). Further, if N , the number of superimposed processes, is itself very large, the distribution of the number of renewals in an interval of length $\mu^* \tau / N$ tends to be distributed as a Poisson distribution with mean τ asymptotically.

In order to prove these results, Cox (1962) utilized the following relation:

$$(1.3) \quad \Psi_e(s, \xi) = 1 + \frac{\xi - 1}{\mu^* s} \Psi_0(s, \xi)$$

where $\Psi_0(s, \xi)$ and $\Psi_e(s, \xi)$ and respectively the L.-S.T. of the generating function of the number of renewals in an interval $(0, t)$ for the ordinary and the equilibrium renewal processes [ξ is the argument for the generating function and s for the L.-S.T.]. (See equation 7, p. 38 Cox (1962); our equation (1.3) above differs slightly from this because we use the L.-S.T. while Cox used only a Laplace transform). From (1.3) Cox and Smith (1954) have also proved, by using a Tauberian argument, that $V_e(t)$, the variance of the number of renewals in $(0, t)$ of an equilibrium renewal process is asymptotically (for large t)

$$(1.4) \quad \frac{C^2 t}{\mu^*} + \frac{3\mu_2^{*2} - 2\mu^* \mu_3^*}{6\mu^{*4}}.$$

Cox and Smith (1954) have considered the behaviour of $V_e(t)$ for small t also and have shown that

$$(1.5) \quad V_e(t) = \frac{t}{\mu^*} + o(t).$$

2. Markov Renewal Processes

Let a system make transitions from state to state according to a Markovian transition probability matrix

$$(2.1) \quad P_0 = [p_{ij}] \quad (i, j = 1, 2, \dots, m)$$

($\sum_{j=1}^m p_{ij} = 1, p_{ij} \geq 0$; i refers to the previous state and j to the subsequent one). We further assume that the c.d.f. of the time taken to make the transition from i to j is $F_{ij}(x)$. (See Pyke (1961b).) Let Z_t denote the state of the system at time t and let $N_j(t)$ denote the number of times the state j is visited in the interval $(0, t)$. The process $\{N_j(t); j = 1, 2, \dots, m\}$ is then called a Markov Renewal Process (M.R.P.). If such a process is observed after a long time has elapsed, it will be called an equilibrium M.R.P. (denoted hereafter by E.M.R.P.). Let $\mu_{ij}(r)$ be the r th moment (about zero) of the c.d.f. $F_{ij}(x)$ and for convenience write $\mu_{ij}(1)$ as only μ_{ij} .

The behaviour of an M.R.P. is governed by the $m \times m$ transition distribution matrix

$$(2.2) \quad Q(x) = [Q_{ij}(x)] = [p_{ij}F_{ij}(x)]$$

and we denote, as stated earlier, by $q_{ij}(s)$ the L.S.T. of $Q_{ij}(x)$ and the matrix $[q_{ij}(s)]$ by $q(s)$. The expected value of the number of renewals $N_j(t)$, conditional on the initial state of the process being i (i.e. $Z_0 = i$) is denoted by $M_{ij}(t)$ and its L.S.T. by $m_{ij}(s)$. It has been shown (See Pyke (1961b)) that

$$(2.3) \quad m(s) = (I - q(s))^{-1} - I$$

and $m(s)$ is called the matrix renewal function and plays an important role in the theory of M.R.P. We plan to consider the pooled output when N independent E.M.R.P.'s are superimposed.

3. The generating Function of the Distribution of the Number of Renewals in an E.M.R.P.

Let $\Psi_i^0(s; \xi_1, \dots, \xi_m)$ be the L.S.T. of the generating function of the probabilities (conditional on $Z_0 = i$)

$$(3.1) \quad \text{Prob} \{N_j(t) = n_j; j=1, \dots, m \mid Z_0 = i\}.$$

for an ordinary M.R.P. Let $\Psi_i^e(s; \xi_1, \dots, \xi_m)$ abbreviated later to Ψ_i^e , be the corresponding quantity for an E.M.R.P. It has been shown by Kshirsagar and Gupta (1969) that the vector $\underline{\Psi}^0$, whose i -th element is $\Psi_i^0(s; \xi_1, \dots, \xi_m)$, is

$$(3.2) \quad \underline{\Psi}^0 = (I - q(s) \text{diag}(\xi_1, \dots, \xi_m))^{-1} (I - q(s)) \underline{e}.$$

where

$$(3.3) \quad \text{diag}(\xi_1, \dots, \xi_m) = \begin{bmatrix} \xi_1 & & \\ & \ddots & \\ & & \xi_m \end{bmatrix}$$

$$(3.4) \quad \underline{e}' = [1, \dots, 1].$$

Further, if $N_j(t_0, t)$ denotes the number of visits to state j in the interval $(t_0, t_0 + t)$ of an ordinary M.R.P., it is shown by Kshirsagar and Gupta (1970) that the L.S.T. (w.r.t. t) of the generating function of the probabilities (similar to (3.1)) of $N_j(t_0, t)$ ($j = 1, \dots, m$) conditional on $Z_0 = i$ is

$$(3.5) \quad \sum_{j=1}^m \sum_{k=1}^m \bar{q}_{jk}(t_0, s | Z_0 = i) \xi_k \Psi_k^0(s; \xi_1, \dots, \xi_m) + \{1 - \sum_{j=1}^m \sum_{k=1}^m \bar{q}_{jk}(t_0, s | Z_0 = i)\}$$

where

$$(3.6) \quad \bar{q}_{jk}(t_0, t | Z_0 = i) = \text{Prob} \{Z_{t_0} = j, \text{ the next transition is to state } k, \text{ the time taken for this transition measured from } t_0 \leq t | Z_0 = i\}$$

and $\bar{q}_{jk}(t_0, s | Z_0 = i)$ is its L.S.T. If we now let t_0 tend to infinity in this, we shall get a relation between the L.S.T.'s of the probability generating functions of the $N_j(t)$'s of an E.M.R.P. and an M.R.P. For this one needs to note that

$$(3.7) \quad \bar{q}_{jk}(t_0, s | Z_0 = i) \rightarrow \frac{1}{\mu_{jj}^*} \frac{(p_{jk} - q_{jk}(s))}{s}$$

as $t_0 \rightarrow \infty$, where μ_{jj}^* is the mean of the time interval between two successive visits to state j which will be assumed to have d.f. $G_{jj}(x)$. Using (3.7) and (3.5) we find

$$(3.8) \quad \Psi_i^e = \frac{1}{s} \underline{e}' D_{\mu^*}^{-1} (P_0 - q(s)) \text{diag}(\xi_1, \dots, \xi_m) \underline{\Psi}^0 + 1 - \frac{1}{s} \underline{e}' D_{\mu^*}^{-1} (P_0 - q(s)) \underline{e}$$

where D_{μ^*} is a diagonal matrix, the diagonal elements of which are μ_{jj}^* ($j=1, \dots, m$). On

using (3.2) and the fact that $P_0 \underline{e} = \underline{e}$, (3.8) simplifies to

$$(3.9) \quad \Psi_i^e = 1 + \frac{1}{s} e' D_{\mu^*}^{-1} [P_0 \text{diag}(\xi_1, \dots, \xi_m) - I] \Psi^0$$

Let $\underline{\Psi}^e$ be the vector whose i th element is $\Psi_i^e(s; \xi_1, \dots, \xi_m)$.

This is a generalization of Cox's result (1.3) for renewal processes.

Now denote by $\underline{N}^O(t)$ the vector of the number of visits to the different states in the ordinary M.R.P., and by $\underline{N}^e(t)$ the same vector for the equilibrium M.R.P. By taking the inverse transform of (3.9) and comparing coefficients of $\xi_1^{n_1} \xi_2^{n_2} \dots \xi_m^{n_m}$, the probabilities of $\underline{N}^e(t)$ may be found in terms of those for $\underline{N}^O(t)$. Define $\underline{e}_k$ to be the m -component vector with 1 in the k th position and zero elsewhere. Let $\underline{n} = [n_1, \dots, n_m]'$, and let $n_{i_1}, \dots, n_{i_q}$ be the non-zero components of $\underline{n}$. For additional convenience of notation, define

$$P_{\underline{n}}^u(t|i) = \text{Prob}(\underline{N}^u(t) = \underline{n} | Z_0 = i)$$

for $u = o, e$. Then, if $\underline{n} \neq \underline{0}$,

$$(3.10) \quad P_{\underline{n}}^e(t|i) = \sum_{j=1}^m \sum_{k=1}^q \frac{1}{\mu_{jj}^*} \int_0^t [p_{jn_k} P_{\underline{n}-\underline{e}_{n_k}}^o(x) - \frac{1}{\lambda} P_{\underline{n}}^o(x)] dx$$

and, if $\underline{n} = \underline{0}$,

$$(3.11) \quad P_{\underline{0}}^e(t|i) = 1 - \sum_{j=1}^m \frac{1}{\mu_{jj}^*} \int_0^t P_{\underline{0}}^o(x) dx.$$

It should be noted that, in the particular case $m=1$, (3.10) reduces to the result (13) (page 94) of Cox and Smith (1954).

4. Superposition of N E.M.R.P.'s

Consider now the pooled output when N independent E.M.R.P.'s, each governed by the same transition distribution matrix $Q(x)$, are superimposed.

Let us first find the variance of the total number of visits to state j in the pooled output.

One way of doing this is to put $\xi_1 = 1, \xi_2 = 1, \dots, \xi_{j-1} = 1, \xi_{j+1} = 1, \dots, \xi_m = 1$ in (3.9) and get a relation between the L.S.T.'s of the p.g.f.'s of $N_j(t)$ in an ordinary and equilibrium M.R.P. Then we can differentiate this relation twice with respect to ξ_j and

obtain a relation between the second factorial moments of the $N_i(t)$'s in an ordinary and equilibrium process. We can further use the fact (proved by Kshirsagar and Gupta (1969)) that the L.-S.T. of the second factorial moment of $N_j(t)$ (conditional on $Z_0 = i$) in an ordinary M.R.P. is the element in the i -th row and j -th column of the matrix $\underline{r}(s) = [r_{ij}(s)]$ where

$$(4.1) \quad \underline{r}(s) = 2 m(s) \text{diag} (\dots m_{jj}(s) \dots),$$

$m_{jj}(s)$ being the j -th diagonal element of $m(s)$, defined in (2.3). This method will be analogous to Cox's method of deriving the variance of the number of renewals in an equilibrium renewal process.

However, there is a better and simpler method of obtaining the variance of the $N_j(t)$'s in an E.M.R.P. For this it is necessary to observe that, if we concentrate only on the visits to a particular state, say j , of an M.R.P., and ignore all the other states, then the resulting process is an ordinary renewal process (as considered in section 1; see Cinlar (1975)), and the interval between two successive visits to state j of this derived renewal process has the c.d.f. $G_{jj}(x)$. (Pyke (1961b) calls it the recurrence time of state j .) The L.-S.T. of $G_{jj}(x)$, namely $g_{jj}(s)$, is given by

$$(4.2) \quad g_{jj}(s) = \frac{m_{jj}(s)}{1 + m_{jj}(s)}$$

where $m_{jj}(s)$ is already defined in (2.3). Let us denote the r th moment (about zero) of the c.d.f. $G_{jj}(x)$ by $\mu_{jj}^*(r)$ ($r = 1, 2, \dots$) where the index r may be omitted when $r = 1$.

We are, therefore, in a position to apply Cox and Smith's results, stated in section 1, for the superposition of equilibrium renewal processes, to this situation. We need simply to replace $G(x)$ in section 1 by $G_{jj}(x)$. Thus, from (1.2), the interval between successive visits to state j in the pooled output of N independent E.M.R.P.'s has the p.d.f.

$$(4.3) \quad -\frac{d}{dy} \left\{ (1 - G_{jj}(y)) \left[\int_y^\infty \frac{1 - G_{jj}(x)}{\mu_{jj}^*(1)} dx \right]^{N-1} \right\}.$$

Similarly, by using (1.5), we find the asymptotic variance of $N_j^P(t)$, the number of visits to state j in an interval $(0, t)$ in the pooled output of N independent E.M.R.P.'s, is

$$(4.4) \quad N \left[\frac{C_{jj}^2 t}{\mu_{jj}^*(1)} + \frac{3 \mu_{jj}^{*2}(2) - 2 \mu_{jj}^*(1) \mu_{jj}^*(3)}{6 \mu_{jj}^{*4}(1)} \right]$$

where C_{jj}^* is the coefficient of variation and $\mu_{jj}^*(r)$ is the r th moment of the c.d.f. $G_{jj}(x)$. However, these are not yet known in terms of the parameters of the M.R.P.'s, namely $[p_{ij}]$ and $[F_{ij}]$. It is, therefore, necessary to do so. This is done in the next section.

Before that, we also observe, from section 1, that if N , the number of E.M.R.P.'s superimposed, is very large, then the distribution of $N_j^P(t)$, where $\tau = tN/\mu_{jj}^*(1)$, tends to a Poisson distribution with mean τ .

5. The Moments of $G_{jj}(x)$

To use (4.4) we need the first three moments of $G_{jj}(x)$. Obviously $\mu_{jj}^*(r)$ is the coefficient of $(-s)^r/r!$ in $g_{jj}(s)$, the L.-S.T. of $G_{jj}(x)$. To expand $g_{jj}(s)$, as given in (4.2), we need to expand $m_{jj}(s)$ in powers of s . For this we use the expansion

$$\begin{aligned} (I - q(s))^{-1} &= I + m(s) \\ (5.1) \quad &= \frac{1}{s} A_{-1} + A_0 + sA_1 + s^2 A_2 + \dots \end{aligned}$$

of the matrix renewal function $m(s)$, given by Kshirsagar and Gupta (1972). $A_{-1}, A_0, A_1, \dots$ are $m \times m$ matrices defined as below:

$$(5.2) \quad A_{-1} = \frac{1}{k_1} L = \frac{1}{k_1} \underline{e} \underline{u}',$$

$$(5.3) \quad A_0 = \left(I - \frac{1}{k_1} LP_1 \right) Z \left(I - \frac{1}{k_1} P_1 L \right) + \frac{k_2}{2k_1^2} L,$$

$$\begin{aligned} (5.4) \quad A_1 &= \left\{ -ZP_1 + \frac{1}{k_1} kP_1 ZP_1 + \frac{1}{2k_1} LP_2 \right\} A_0 \\ &+ \frac{1}{2k_1} \left\{ ZP_2 - \frac{1}{k_1} LP_1 ZP_2 - \frac{1}{3k_1} LP_3 \right\} L \end{aligned}$$

where $\underline{e}$ is the $m \times 1$ column vector of unit elements and $\underline{u}$ is the vector of the stationary state probabilities of the imbedded Markov Chain with P_0 as the transition probability matrix, i.e.,

$$(5.5) \quad \underline{u}' P_0 = \underline{u}'$$

In addition,

$$(5.6) \quad Z = (I - P_0 + L)^{-1},$$

$$(5.7) \quad P_k = \int_0^\infty x^k dQ(x) \quad (k = 0, 1, 2, \dots),$$

$$(5.8) \quad k_r = \underline{u}' P_r \underline{e} \quad (r = 1, 2).$$

Let us denote by $a_{ij}(k)$ ($k = -1, 0, 1, \dots$) the element in the i -th row and j -th column of A_k . Using the expansion of $m(s)$ in (4.2) and expanding it further in powers of s , we obtain, after some algebra

$$(5.9) \quad \mu_{ij}^*(1) = \frac{1}{a_{ij}(-1)},$$

$$(5.10) \quad \mu_{ij}^*(2) = \frac{2a_{ij}(0)}{a_{ij}^2(-1)},$$

and

$$(5.11) \quad \mu_{ij}^*(3) = 6 \left[\frac{a_{ij}^2(0)}{a_{ij}^3(-1)} - \frac{a_{ij}(1)}{a_{ij}^2(-1)} \right].$$

The variance σ_{ij}^{*2} and the coefficient of variation C_{ij}^* of $G_{ij}(x)$ can be obtained from these moments. Thus

$$(5.12) \quad \sigma_{ij}^{*2} = \frac{2 a_{ij}(0) - 1}{a_{ij}^2(-1)}$$

and

$$(5.13) \quad C_{ij}^{*2} = 2 a_{ij}(0) - 1.$$

Using these expressions in (4.4), we find the asymptotic result,

$$(5.14) \quad V \{N_j^P(t)\} \sim N[a_{ij}(-1)(2 a_{ij}(0) - 1)t + 2 a_{ij}(-1) a_{ij}(1)]$$

for large t .

Similarly, using (1.6), for small t , it can be readily seen that

$$(5.15) \quad V \{N_j^P(t)\} \text{ is approximately } = t \lambda_{ij}^*$$

where

$$(5.16) \quad \lambda_{jj}^* = \frac{N}{\mu_{jj}^*(1)} = N a_{jj} (-1)$$

Cox and Smith have given more precise results for the particular case when the distribution of the interval between two successive renewals is a χ^2 with 2ν degrees of freedom. We can use this if the recurrence time distribution $G_{jj}(x)$ is of the χ^2 type with 2ν d.f. Then

$$(5.17) \quad V \{N_j^P(t)\} \sim N\left[\frac{t}{2\nu^2} + \frac{1}{6}\left(1 - \frac{1}{\nu^2}\right)\right]$$

as $t \rightarrow \infty$. Also, when $2\nu = 6$, a simple expression for the variance is

$$(5.18) \quad N\left[\frac{t}{9} + \frac{4}{27}\left(1 - e^{-\frac{3}{2}t} \cos \frac{\sqrt{3}t}{2}\right)\right].$$

From (5.17) it can be seen that, if we plot $V \{N_j^P(t)\}$ against t , the intercept of the asymptote is

$$(5.19) \quad \frac{N}{6}(1 - C_{jj}^{*4})$$

as $C_{jj}^{*2} = 1/\nu$ for a χ^2 distribution with 2ν d.f. Cox and Smith utilize these results for estimating N and C_{jj}^{*2} from the results of the pooled output, and they can be used for E.M.R.P.'s also in exactly the same way.

Downton (1972) also has considered the problem of the superposition of renewal processes in some particular cases. His results can also be utilized in our problem of the superposition of Markov renewal processes.

If the L.-S.T. of the recurrence time distribution for the j -th state, namely $g_{jj}(s)$, is of the form

$$(5.20) \quad \frac{\lambda_1}{\lambda_1 + s} \cdot \frac{\lambda_2}{\lambda_2 + s},$$

Downton has shown that the time to the first visit to state j , when N identical M.R.P.'s (not necessarily in equilibrium) are superimposed, will have the c.d.f. (when $\lambda_2 \rightarrow \lambda_1$ in the limit)

$$(5.21) \quad 1 - (1 + \lambda_1 t)^N e^{-N\lambda_1 t}$$

He further shows that for large N this distribution can be approximated by a Weibull distribution. This raises the question of what the transition c.d.f.'s $F_{ij}(x)$, must be to yield $g_{ij}(s)$ of the form (5.20). This is being investigated.

Downton gives similar results about the distribution function of the interval associated with the $(m+1)$ th visit to state j in the superposed process. He also considers the asymptotic behaviour of the number of visits to state j in the pooled output, and it is not surprising that this is a Poisson distribution, as we have already seen.

6. The recurrence time distribution

We saw in the last section that if $G_{ij}(x)$ is the d.f. of a χ^2 with 2ν d.f., we get exact or more precise results for the variance of the number of visits to any particular state in the superposed process when N independent E.M.R.P.'s are pooled. This raises the interesting question of what the original transition distributions $F_{ij}(x)$ should be to yield $G_{ij}(x)$ to be the c.d.f. of a χ^2 distribution.

In general it is not always possible to derive transition d.f. $F_{ij}(x)$ which yield arbitrary χ^2 distribution functions $G_{ij}(x)$. Whether or not such F_{ij} exist depends on the degrees of freedom of the χ^2 d.f. For example, suppose the number of states in the MRP under consideration is $M = 2$ and assume $G_{ij}(x)$ is a χ^2 -distribution with $2k$ d.f. for $i, j = 1, 2$. Then it can be shown that the desired F_{ij} are exponential distribution functions if $k=1$. On the other hand, no distribution F_{ij} will yield the G_{ij} if $k=2$.

7. Application to a Human Reproduction Model

In this section, we now give an application of our results to a human reproduction model considered by Perrin and Sheps (1964). They consider the reproductive pattern of a married female known to be non-pregnant and fecundable at the time of marriage. At any time after marriage this woman can be assumed to be in one and only one of the following states:

- S_1 = non-pregnant, fecundable state,
- S_2 = pregnant state,
- S_3 = postpartum sterile period associated with abortion or fetal loss,
- S_4 = postpartum sterile period associated with still birth,
- S_5 = postpartum sterile period associated with live birth.

The matrix of transition distributions assumed by them is

$$(7.1) \quad Q = [Q_{ij}(x)] = \begin{bmatrix} 0 & \rho \sum_{t=0}^{\infty} (1-\rho)^t & 0 & 0 & 0 \\ 0 & 0 & p_{23}F_{23}(x)p_{24}F_{24}(x)p_{25}F_{25}(x) & & \\ F_{31}(x) & 0 & 0 & 0 & 0 \\ F_{41}(x) & 0 & 0 & 0 & 0 \\ F_{51}(x) & 0 & 0 & 0 & 0 \end{bmatrix}$$

This model assumes that the probability of pregnancy is a constant ρ with waiting time in the non-pregnant state distributed geometrically. Pregnancy terminates in either fetal death, stillbirth, or live birth with probability p_{23} , p_{24} , or p_{25} respectively, and the variable period of gestation is distributed according to F_{23} , F_{24} , or F_{25} which may depend on the manner in which pregnancy terminates. A variable period of infecundability follows with its length again depending on the form of pregnancy termination according to the c.d.f.'s F_{31} , F_{41} , and F_{51} . Transition to the non-pregnant, fecundable state follows, and so p_{31} , p_{41} , and p_{51} are all 1 in this model.

From (7.1) it can be shown that $(I - q(s))^{-1}$ is given by

$$(7.2) \quad \frac{1}{\alpha} \begin{bmatrix} 1 & \beta & \beta p_{23}f_{23}(s) & \beta p_{24}f_{24}(s) & \beta p_{25}f_{25}(s) \\ \sum_{k=3}^5 p_{2k}f_{2k}(s)f_{k1}(s) & 1 & p_{23}f_{23}(s) & p_{24}f_{24}(s) & p_{25}f_{25}(s) \\ f_{31}(s) & \beta f_{31}(s) & \alpha + \beta p_{23}f_{23}(s)f_{31}(s) & \beta p_{24}f_{24}(s)f_{31}(s) & \beta p_{25}f_{25}(s)f_{31}(s) \\ f_{41}(s) & \beta f_{41}(s) & \beta p_{23}f_{23}(s)f_{41}(s) & \alpha + \beta p_{24}f_{24}(s)f_{41}(s) & \beta p_{25}f_{25}(s)f_{41}(s) \\ f_{51}(s) & \beta f_{51}(s) & \beta p_{23}f_{23}(s)f_{51}(s) & \beta p_{24}f_{24}(s)f_{51}(s) & \alpha + \beta p_{25}f_{25}(s)f_{51}(s) \end{bmatrix}$$

$$\text{where } \beta = \frac{\rho}{1 - e^{-s}(1-\rho)} \quad \text{and} \quad \alpha = 1 - \beta \sum_{k=3}^5 p_{2k}f_{2k}(s)f_{k1}(s).$$

From (4.2) and (7.2), the L.-S.T. of $G_{ij}(x)$ is obtained. Except for a change of notation, these are the same as the expressions found by Perrin and Sheps. We have, $j = 1$ or 2 ,

$$(7.3) \quad g_{ji}(s) = \rho [1 - e^{-s}(1-\rho)]^{-1} \sum_{k=3}^5 p_{2k}f_{2k}(s)f_{k1}(s) \quad \text{and, for } i = 3, 4, \text{ or } 5,$$

$$(7.4) \quad g_{jj}(s) = \rho p_{2j} f_{2j}(s) f_{j1}(s) \left[1 - \rho \sum_{\substack{k=3 \\ k \neq j}}^5 p_{2k} f_{2k}(s) f_{k1}(s) - e^{-s}(1-\rho) \right]^{-1}$$

The first three moments of $G_{jj}(x)$ are obtained by differentiating $g_{jj}(s)$.

They are

$$(7.5) \quad \mu_{jj}^*(1) = \frac{\delta}{p_{2j}}$$

$$(7.6) \quad \mu_{jj}^*(2) = \frac{\delta(2)}{p_{2j}} + \frac{2\omega_j \delta}{p_{2j}^2}$$

and

$$(7.7) \quad \mu_{jj}^*(3) = \frac{\delta(3)}{p_{2j}} + \frac{3\omega_j(2)\delta}{p_{2j}^2} + \frac{3\omega_j \delta(2)}{p_{2j}^2} + \frac{6\omega_j^2 \delta}{p_{2j}^3}$$

where $p_{21} = p_{22} = 1$, and

$$(7.8) \quad \delta(r) = \rho^{-1} - 1 + \sum_{k=3}^5 p_{2k} \eta_k(r),$$

$$(7.9) \quad \eta_k(r) = (-1)^r \left. \frac{d^r}{ds^r} (f_{2k}(s) f_{k1}(s)) \right|_{s=0}$$

and

$$(7.10) \quad \omega_j(r) = \begin{cases} \rho^{-1} - 1 & \text{for } j = 1, 2 \\ \rho^{-1} - 1 + \sum_{k=3}^5 p_{2k} \eta_k(r) - p_{2j} \eta_j(r) & \text{for } j = 3, 4, 5. \end{cases}$$

The index r is omitted when $r = 1$.

Equations (5.14) – (5.16) then become, respectively,

$$(7.11) \quad V \{N_j^P(t)\} \sim \frac{N p_{2j} t}{\delta^3} [p_{2j} \delta(2) + 2\delta \omega_j - \delta^2] \\ + \frac{N p_{2j}}{6\delta^4} [3p_{2j} \delta^2(2) - 2\delta p_{2j} \delta(3) - 6\delta^2 \omega_1 \delta(2) + 6\delta \omega_j \delta(2)],$$

$$(7.12) \quad V \{N_j^P(t)\} \sim t \lambda_{jj}^*,$$

and

$$(7.13) \quad \frac{1}{\lambda_{jj}^*} = -\frac{\delta}{Np_{2j}},$$

where $j = 1, 2, \dots, 5$.

8. Application to a Predator-Prey Model

Holling (1966) considered the following four states representing the activity of a predator seeking its prey:

S_1 = searching state,

S_2 = pursuit state,

S_3 = handling and eating state, and

S_4 = digestion state.

Rao and Kshirsagar (1978) assume an M.R.P. model for these four states and, using Holling's analysis, consider the following pattern for the transition distribution matrix $[Q_{ij}(x)]$:

$$(8.1) \quad Q = \begin{bmatrix} p_{11}F_{11}(x) & p_{12}F_{12}(x) & 0 & 0 \\ p_{21}F_{21}(x) & 0 & p_{23}F_{23}(x) & 0 \\ p_{31}F_{31}(x) & 0 & 0 & p_{34}F_{34}(x) \\ p_{41}F_{41}(x) & 0 & 0 & 0 \end{bmatrix}$$

From this, they have further shown that the L.-S.T. of the matrix renewal function is, by using (2.3),

$$(8.2) \quad m(s) = \frac{1}{Z} \begin{bmatrix} a+bc+bde+bdfg & b & bd & bdf \\ c+dc+dfg & bc+bdc+bdfg & d(1-a) & (1-a)df \\ e+fg & bc+bf g & bde+bdfg & f(1-a-bc) \\ g & bg & bdg & bdfg \end{bmatrix}$$

where

$$Z = 1 - a - bc - bde - bdfg, \quad a = p_{11}f_{11}(s), \quad b = p_{12}f_{12}(s), \quad c = p_{21}f_{21}(s),$$

$$d = p_{23}f_{23}(s), \quad e = p_{31}f_{31}(s), \quad f = p_{34}f_{34}(s), \quad g = p_{41}f_{41}(s),$$

$$\text{and} \quad f_{ij}(s) = \text{L.-S.T. of } F_{ij}(x).$$

For notational convenience the following conversion is useful. Let θ_k and h_k be defined by (for $k = 1, 2, \dots, 7$),

$$\begin{aligned} \theta_1 &= p_{11}, \theta_2 = p_{12}, \theta_3 = p_{21}, \theta_4 = p_{23}, \theta_5 = p_{31}, \\ \theta_6 &= p_{34}, \theta_7 = p_{41}; \quad h_1(s) = f_{11}(s), h_2(s) = f_{12}(s), \\ (8.3) \quad h_3(s) &= f_{21}(s), h_4(s) = f_{23}(s), h_5(s) = f_{31}(s), \\ h_6(s) &= f_{34}(s), \text{ and } h_7(s) = f_{41}(s). \end{aligned}$$

Furthermore, for $r = 1, 2$, and 3 , let

$$(8.4) \quad \beta_r(s) = \theta_1 h_1^r(s) + \sum_{\ell=1}^r \left\{ \theta_{2\ell+1} h_{2\ell+1}(s) \prod_{k=1}^{\ell} \theta_{2k} h_{2k}(s) \right\}$$

and let $\beta_0(s) = \theta_1 h_1(s)$ and $\beta_{-1}(s) = 0$.

It can then be shown from (4.2) and (8.2) that the L.-S.T. of the distribution of recurrence time of state i is given by

$$(8.5) \quad g_{ii}(s) = \frac{\beta_3(s) - \beta_{i-2}(s)}{1 - \beta_{i-2}(s)}$$

for $i = 1, 2, 3$, or 4 . By differentiating $g_{ii}(s)$, the first three moments of $G_{ij}(x)$ are seen to be

$$(8.6) \quad \mu_{ii}^* = \frac{B_1(i)}{A_1^2(1-2)}$$

$$(8.7) \quad \mu_{ii}^*(2) = \frac{B_2(i)}{A_1^2(i-2)} + \frac{2A_2(i-2)B_1(i)}{A_1^3(i-2)}$$

and

$$(8.8) \quad \mu_{ii}^*(3) = \frac{B_3(i)}{A_1^2(i-2)} + \frac{6A_2^2(i-2)B_1(i)}{A_1^4(i-2)} + \frac{3A_2(i-2)B_2(i)}{A_1^3(i-2)} + \frac{3A_3(i-2)B_1(i)}{A_1^3(i-2)}$$

In these expressions,

$$(8.9) \quad B_{k-1}(i) = A_1(i-2)A_k(3) - A_1(3)A_k(i-2)$$

for $k = 2, 3$, or 4 , where

$$(8.10) \quad A_1(r) = 1 - \beta_r(0), \text{ and} \\ A_{k+1}(r) = (-1)^k \left. \frac{d^k \beta_r(s)}{ds^k} \right|_{s=0}$$

for $k = 1, 2$, or 3 . Expressions have been derived which give $A_{k+1}(r)$ explicitly in terms of the moments of $F_{ij}(x)$. For this example, it is again convenient to change the notation for these moments. Let γ_k be the first moment (about zero) corresponding to h_k . Thus, for example, h_1 is the L.-S.T. of $F_{11}(x)$ and so $\gamma_1 = \mu_{11}$. Similarly, denote by $\gamma_k(r)$ the r th moment (about zero) of the distribution whose L.-S.T. is h_k , and let δ_k^2 be the variance of that distribution (thus, $\delta_1^2 = \sigma_{11}^2$ and $\delta_2^2 = \sigma_{12}^2$). Then we have the following:

$$(8.11) \quad A_1(0) = 1 - \theta_1, A_1(-1) = 1, \text{ and for } r = 1, 2, \text{ or } 3, \\ A_1(r) = 1 - \theta_1 - \sum_{\ell=1}^r [\theta_{2\ell+1} \prod_{k=1}^{\ell} \theta_{2k}],$$

$$(8.12) \quad A_2(0) = \theta_1 \gamma_1, A_2(-1) = 0, \text{ and for } r = 1, 2, \text{ or } 3, \\ A_2(r) = \theta_1 \gamma_1 + \sum_{\ell=1}^r [\theta_{2\ell+1} (\gamma_{2\ell+1} + \sum_{k=1}^{\ell} \gamma_{2k}) \prod_{k=1}^{\ell} \theta_{2k}]$$

$$(8.13) \quad A_3(0) = \theta_1 \gamma_1(2), A_3(-1) = 0, \text{ and for } r = 1, 2, \text{ or } 3, \\ A_3(r) = \theta_1 \gamma_1(2) + \sum_{\ell=1}^r \{ \theta_{2\ell+1} [\gamma_{2\ell+1} (2) + 2\gamma_{2\ell+1} \sum_{k=1}^{\ell} \gamma_{2k} + \sum_{k=1}^{\ell} \delta_{2k}^2 \\ + (\sum_{k=1}^{\ell} \gamma_{2k})^2] \prod_{k=1}^{\ell} \theta_{2k} \},$$

and $A_4(0) = \theta_1 \gamma_1(3)$, $A_4(-1) = 0$, and for $r = 1, 2$, or 3 ,

$$\begin{aligned}
 A_4(r) = & \theta_1 \gamma_1(3) + \sum_{\ell=1}^r \{ \theta_{2\ell+1} [\gamma_{2\ell+1}(3) + 3\gamma_{2\ell+1}(2)] \sum_{k=1}^{\ell} \gamma_{2k} \\
 & + 3\gamma_{2\ell+1} \left(\sum_{k=1}^{\ell} \gamma_{2k} \right)^2 + \left(\sum_{k=1}^{\ell} \gamma_{2k} \right)^3 + 3(\gamma_{2\ell+1} + \sum_{k=1}^{\ell} \gamma_{2k}) \sum_{k=1}^{\ell} \delta_{2k}^2 \\
 & + \sum_{k=1}^{\ell} (\gamma_{2k}(3) - 3\gamma_{2k}(2) + 2\gamma_{2k}^3) \prod_{k=1}^{\ell} \theta_{2k} \} .
 \end{aligned}
 \tag{8.14}$$

From μ_{ii}^* , $\mu_{ii}^*(2)$, and $\mu_{ii}^*(3)$, expressions are obtained for the results (5.14), (5.15), and (5.16). The case when $i=1$ (where the predator is in the searching state) reduces to simpler form since it is easily seen that

$$\begin{aligned}
 \mu_{11}^* &= A_2(3) = A_2, \\
 \mu_{11}^*(2) &= A_3(3) = A_3, \text{ and} \\
 \mu_{11}^*(3) &= A_4(3) = A_4
 \end{aligned}
 \tag{8.15}$$

where the index 3 will be omitted with no confusion. Then (5.14), (5.15), and (5.16) are, respectively,

$$V \{N_1^p(t)\} \sim \frac{N(A_3 - A_2^2)t}{A_2^3} + \frac{N(3A_3^2 - 2A_2A_4)}{6A_2^4},
 \tag{8.16}$$

$$V \{N_1^p(t)\} \sim \lambda_{11}^* t
 \tag{8.17}$$

$$\text{where } \frac{1}{\lambda_{11}^*} = \frac{A_2}{N}.
 \tag{8.18}$$

The results of this section extend in a straight forward manner to any M.R.P. of m states in which the L.S.T. of the Q-matrix is of the form

$$(8.19) \quad q(s) = \begin{bmatrix} \theta_1 h_1(s) & \theta_2 h_2(s) & 0 & \dots & 0 \\ \theta_3 h_3(s) & 0 & \theta_4 h_4(s) & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \theta_{2m-3} h_{2m-3}(s) & 0 & 0 & \dots & \theta_{2m-2} h_{2m-2}(s) \\ \theta_{2m-1} h_{2m-1}(s) & 0 & 0 & \dots & 0 \end{bmatrix}$$

Equation (8.5) for $g_{ij}(s)$ becomes

$$(8.20) \quad g_{ij}(s) = \frac{\beta_{m-1}(s) - \beta_{i-2}(s)}{1 - \beta_{i-2}(s)}$$

for $i = 1, 2, \dots, m$ where $\beta_i(s)$ is still given by (8.4) but now $r = 1, 2, \dots, m-1$.

9. Superposition of Non-identical M.R.P.'s

Downton (1972) has referred to Drenick's (1960) generalization of the Cox-Smith result (1954) to superposition of n independent non-identical ordinary renewal processes. These can be applied to consider the superposition of n independent M.R.P.'s in the same way as identical M.R.P.'s. The conditions under which Drenick's result of asymptotic Poisson distribution of the number of renewals in the superposed process holds are as follows: Let μ_i be the mean of the c.d.f. $F_i(t)$ of the interval between successive renewals for the i th process, and let $\bar{\mu}_n$ be the reciprocal of $\sum_{i=1}^n (1/\mu_i)$.

Then we need

$$(i) \quad \lim_{n \rightarrow \infty} \sup_i (\bar{\mu}_n / \mu_i) = 0$$

$$(ii) \quad F_i(t) \leq a_i t^\delta \text{ as } t \rightarrow 0 \text{ } (a_i > 0, \delta > 0)$$

$$\text{and} \quad (iii) \quad a_i \leq A.$$

Obviously, when we apply this result to M.R.P.'s, we will need to replace μ_i above by the mean of the recurrence time distribution G_{ij} for the i -th M.R.P., and F_i by the c.d.f. G_{ij} of the i -th process. However, unless these conditions can be translated into the conditions on the c.d.f.'s $Q_{ij}(x)$ of the matrix Q for the i -th process, the result would not be

of much value.

Downton (1972) also refers to results of Grigelionis (1964).

10. Concluding Remarks

In summary, this paper gives results about the total number of visits to any particular state in the pooled output of n independent identical M.R.P.'s and also about the distribution of the interval between two successive visits to the same state. However, we have not been successful as yet in deriving the distribution of the interval between a visit to the i -th state and a visit to the j -th state ($i \neq j$) thereafter in the pooled output.

REFERENCES

- CINLAR, E. (1975), *Introduction to Stochastic Processes*, Englewood Cliffs; Prentice-Hall.
- COX, D.R. (1962), *Renewal Theory*, London: Methuen.
- COX, D.R. and SMITH, W.L. (1954), "On the superposition of renewal processes," *Biometrika*, **41**, 91.
- DOWNTON, F. (1972), "Stochastic models for successive failures," *Int. Statist. I. (B)*, **44**, 677.
- DRENICK, R.F. (1960), "The failure law of complex equipment", *J. Soc. Indust. Appl. Math.*, **8**, 680.
- GRIGELIONIS, B.I. (1964), "Limit theorems for sums of renewal processes," *Cybernetics in the Service of Communism*, **2**, *Collected Articles in Reliability and Queuing Theory*, 246, Moscow-Leningrad: Izdatel'stvo Emergia.
- HOLLING, C.S. (1966), "The functional response of invertebrate predators to prey density," *Mem. Ent. Soc. Can.*, **48**, 3.
- KSHIRSAGAR, A.M. and GUPTA, Y.P. (1969) "Some results in Markov renewal processes," *Cal. Statist. Ass. Bull.*, **18**, 61.
- KSHIRSAGAR, A.M. and GUPTA, Y.P. (1970), "Distribution of the number of Markovian renewals in an arbitrary interval," *Austral. J. Statist.*, **12** (1), 58.
- KSHIRSAGAR, A.M. and GUPTA, Y.P. (1972), "A note on the matrix renewal function" *Austral. J. Statist.*, **13**(4), 417.
- PERRIN, E.B. and SHEPS, M.C. (1964), "Human reproduction: a stochastic process," *Biometrics*, **20**, 28.
- PYKE, R. (1961a), "Markov renewal processes: definitions and preliminary properties," *Ann. Math. Statist.*, **32**, 1231.
- PYKE, R. (1961b), "Markov renewal processes with finitely many states," *Ann. Math. Statist.*, **32**, 1243.
- RAO, C.R. and KSHIRSAGAR, A.M. (1978), "A semi-Markovian model for predator-prey interactions," *Biometrics*, **34**, 611.